



Quality Assessment of Estimates for Decision-Making

Module 2 - Session 2.5.1

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Question today...

When we derive indicators from mobile phone data, how can we assess whether they are reliable enough to support decision-making?

Contents

Dimensions of quality assessment for aggregates and estimates derived from mobile phone data

- Completeness, coverage, and bias: identifying gaps and representation issues
- Plausibility, temporal consistency, and sensitivity to assumptions. Validation using external benchmarks and reference data.
- Communicating uncertainty and assessing fitness for use

Learning objectives

1. Describe key dimensions of quality assessment
2. Identify coverage and representativeness issues
3. Conduct plausibility checks and validate with external data
4. Communicate limitations and assess fitness for policy/statistical use

Dimensions of quality assessment

Why quality assessment of aggregates is important?

- Mobile phone data is a not random sample and not collected for producing statistics.
- There are risks of misinterpretation due to:
 - Incomplete coverage of target population or event
 - Biased patterns
 - Misleading patterns



Without proper quality assessment, decision-makers may draw misleading conclusions.

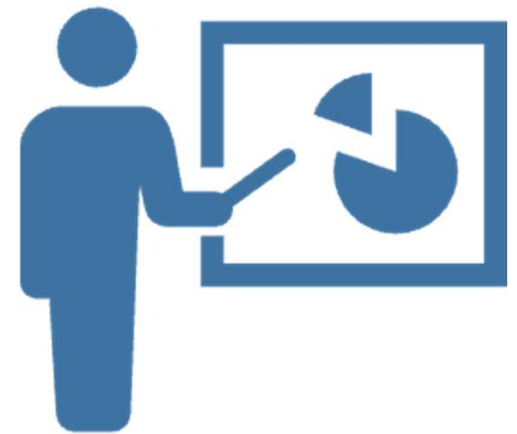
Quick poll I

Question

When you think about using mobile phone data in your organization, which quality issue worries you most?

Options

- A. Coverage – some people or areas are missing
- B. Bias – some groups are over- or under-represented
- C. Timeliness – data are not up-to-date enough
- D. Communicating limitations to decision-makers



Three dimensions of quality assessment

Dimension 1: Data integrity & coverage	Dimension 2: Analytical robustness	Dimension 3: Interpretability & use
• Completeness • Coverage • Bias assessment	• Plausibility • Temporal consistency • Sensitivity • Validation	• Communication of limits • Fit for purpose

Dimension I

Integrity & coverage

Dimension I – Integrity & coverage

Key questions:

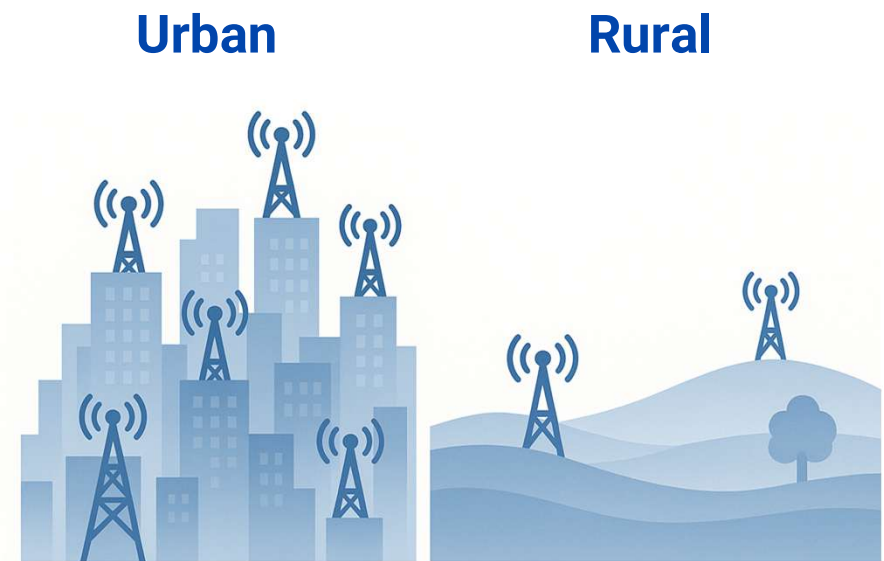
- Are the aggregate values complete over time and space?
- Do they reflect the intended population or geographic area?



Coverage gaps

Example: Regional inconsistencies in aggregate availability due to tower density

- Rural areas often have fewer cell towers, which reduces the volume and quality of data that can be aggregated.
- As a result, the extracted patterns tend to underestimate the actual conditions observed on the ground.



Use case: Estimating commuting population in urban and rural areas

Context: Transport Authority is using MPD to **estimate daily commuting flows** within local districts, with the aim of informing intra-district transport planning

Region A – Urban District

- Dense cell towers
- Short commuting trips



Region B – Rural District

- Sparse cell towers
- Short commuting trips



Use case: Estimating commuting population in urban and rural areas

Region B – Rural District

Commuting within a rural town, or between nearby villages, is often not detected as a separate trip.



For Region B (Rural District), the actual number is expected to be much higher. The aggregates can be adjusted using school enrollment and labor force survey data.

Quick poll 2

Question

If rural commuting flows are under-detected because of sparse tower coverage, what is the **most likely consequence for investment decision**?

Options

- A. Rural areas receive too many transport investments
- B. Rural areas receive too few transport investments
- C. No impact, because planners will ignore the data



Dimension 2

Analytical robustness

Dimension 2 – Analytical robustness

Key questions

- How sensitive are the indicators to changes in modeling assumptions, thresholds, or processing methods?
- Are the indicator values plausible and consistent with known trends or events?
- Do alternative methods or data sources produce similar results?



Sensitivity to analytical assumptions

Overview

- Indicators rely on analytical assumptions and parameter definitions (e.g., user activity thresholds, trip distance cutoffs)
- A clear understanding of how the data are generated is critical for assessing the validity and robustness of the indicator.
- Clear documentation improves reproducibility and supports methodological rigor.

Example: Scale with assumption

Context: Census data is often used to construct scaling factors to adjust for representativeness in mobile phone data. However, censuses **may undercount hard-to-reach populations**, resulting in misleading conclusions.

District A – Residential area

- Formal housing, stable population
- Census coverage is relatively complete
- Scaling factor aligns well with ground truth

District B – Informal-settlement area

- High share of seasonal workers and informal dwellers
- Census undercounts the segments of the population
- Scaling factor underestimates actual population presence

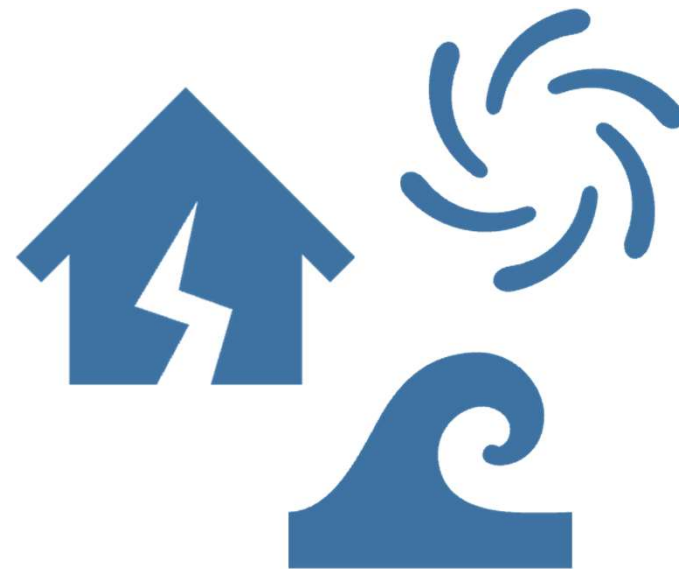
Use case: Utilize dynamic population estimates for disaster response

Authorities want to use population estimates from MPD to support emergency supply distribution during a disaster.



In District B, where informal settlements are concentrated:

- The scaled population estimate may fall below the actual population.
- As a result, emergency supplies could be under-allocated.



Use case: Utilize dynamic population estimates for disaster response (cont.)

To address this:

- Consider running multiple scaling scenarios to assess how results vary under different assumptions.
- If undercoverage is known, adjust the scaling factors based on local context. For example:
 - Compare home locations across seasons to detect population shifts and consider applying season-based scaling factors by season.
 - Alternatively, flag such areas as low-confidence zones.

Quick poll 3

Question

If census data undercount informal settlements and we use those census counts to scale CDR estimates, what is the most likely effect on the scaled population in those areas?

Options

- A. The scaled population will be too high
- B. The scaled population will be too low
- C. Scaling will remove the bias completely
- D. There will be no effect on disaster planning



Check for plausibility and temporal logics

Overview

- **Plausibility checks** help flag values that are too high, too low, or out of pattern.
- **Temporal logic** involves evaluating whether changes over time (e.g., trends, cycles, shocks) make sense in the context of:
 - Weekdays vs. weekends
 - Holidays
 - Seasonality
 - Events



Example I: Confirm weekly cycles match land use patterns

Estimated result: Dynamic population estimate shows a sharp rise in a commercial area on a Sunday.

Context: Weekends are expected to show lower presence in the city center.



Example 1: Confirm weekly cycles match land use patterns

What to check:

- Does the weekend population pattern differ from weekdays as expected?
- Are population values unusually high for a non-workday?

Possible issue/interpretation:

- Error in home location assignment or data gap on previous day.
- An event or anomaly (e.g., market, protest) not captured in the data.



Example 2: Validate seasonal population shifts from migration

Estimated result:

Monthly residential population estimates show an increase in rural districts but the trend does not align with the known harvest season.

Context:

Seasonal migration patterns are often predictable, as they are tied to agricultural cycles. Residential population estimates should reflect these shifts.



Example 2: Validate seasonal population shifts from migration

What to check:

- Are the increases aligned with the agricultural calendar or labor cycles?
- Are origin regions showing expected declines?

Possible issue/interpretation:

- The home detection algorithm may be misclassifying home locations.
- Aggregation over fixed monthly periods may smooth real migration patterns.



Quick poll 4

Question

Your dynamic population estimates show an unexpected spike in the city center on Sundays. What is the most appropriate first step?

Options

- A. Announce that Sunday has become the main workday in the city
- B. Check for data or processing issues and compare with known events
- C. Delete all Sunday data from the series



Validate through comparison and triangulation

Overview

- Estimates should be validated through statistical comparison with external data sources.
- Validation helps identify discrepancies and strengthens confidence in the results.
- Triangulation using multiple sources or methods adds robustness and contextual depth to interpretation.

Example I: Population estimates vs. known population data

How to examine:

Employ regression to evaluate the degree of alignment between population estimates and known population data

What to use as input:

- **Population counts:** best for direct comparisons when spatial units are similar in size.
- **Population density:** useful when comparing areas with different land sizes
- **Log-transformed values:** helps manage extreme differences between small and large populations.

Example I: Population estimates vs. known population data (cont.)

Estimated result: Residential population estimate is 20% higher than known population data in a specific district.

What to check: Is the gap within an explainable range given the local context?

Context: The area is a rapidly growing peri-urban zone near a major city.

Possible issue/interpretation: The difference may reflect recent growth not captured in census data.



Quick poll 5

Question

In a fast-growing peri-urban district, your CDR-based resident population estimate is about 20% higher than the census. What is the best way to proceed?

Options

- A. Assume the census is outdated and use the CDR estimate as the new 'truth'
- B. Discard the CDR indicator because it does not match the census
- C. Investigate possible reasons (growth, mobility patterns, multi-SIM, timing) and document a careful interpretation
- D. Simply reduce the CDR estimate by 20% so it matches the census exactly

Dimension 3

Interpretability & use of indicators

Dimension 3 – Interpretability & use of indicators

Key questions

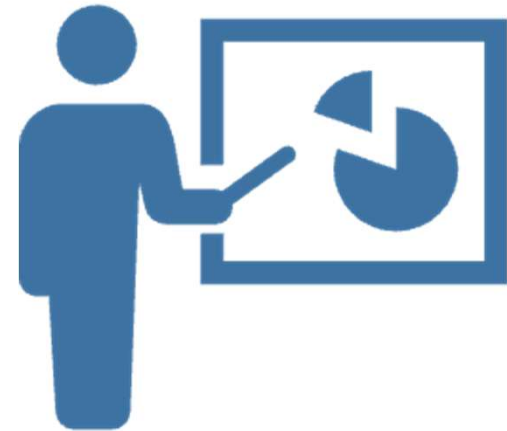
- Are the assumptions, limitations, and sources of uncertainty clearly communicated?
- Is the indicator appropriate for the intended use or decision-making context?



Communicating limitations and uncertainty

- Who is represented?
- Where is coverage weak?
- Key modelling assumptions
- Where is uncertainty higher?

How to communicate the result?



Example: OD matrix scaled using census data without addressing age bias

Estimated results:

- OD matrix generated from MPD, showing movement between districts.
- Scaled using census data to estimate full population-level mobility flows.

Issues

- Population flows captured in the OD matrix reflect relatively mobile segments of the population.
- Children, who are typically less mobile, often make up a large share of the census population, especially in developing countries.
- Scaling with the total census population, including children, assumes equal mobility across all age groups and can lead to misleading results.

Example: OD matrix scaled using census data without addressing age bias (cont.)

Risks

- Travel volumes may be overestimated
- Potential misallocation of resources in planning and intervention

Good practice

- Clearly explain that the OD matrix reflects the mobile, adult population, not the full population
- Flag limitations when using indicators in settings with large youth populations
- If household/travel survey data are available, use age-adjusted scaling factors

Quick poll 6

Question

When you use an OD matrix derived from MPD and scaled with total census population, what does it most realistically represent?

Options

- A. Exact movements of the entire population, including children
- B. Mobility patterns mainly of mobile, adult users, adjusted to approximate the total population
- C. Only tourists and visitors
- D. Only people who use public transport

Introduction to Practical Exercise

Quality assurance: Intro to Practical Exercise

- **Task:** Quality assessment of OD matrices for disaster preparedness

- **Exercise Link:**

<https://github.com/Flowminder/WB-GDF-Modern-Data-Workflows-Code-Tasks>

The screenshot shows the GitHub repository page for 'Flowminder / WB-GDF-Modern-Data-Workflows-Code-Tasks'. The repository is public and has 20 commits. The commit history shows updates to README.md, input data files, and coding exercises. The README section is titled 'WB-GDF: Modern Data Workflows Code Tasks' and describes the repository's purpose. It includes a table of course sessions with links to Colab notebooks.

Session	Session Title	Link to Colab
Session 2.3.4	Input data quality assurance and cleaning	Open in Colab
Session 2.4.1	Continuity models	Open in Colab
Session 2.4.2	Meaningful Locations and Usual Environments	Open in Colab

Exercise: Quality assessment of OD matrices for disaster preparedness

Objective:

Evaluate an Origin–Destination (OD) matrix derived from mobile phone data (MPD) for disaster preparedness.

Contexts:

- A national statistical office is collaborating with emergency response agencies to prepare for disasters, such as floods and cyclones.
- They want to use a MPD-derived OD matrix to understand daily population movements between districts.
- This will inform evacuation planning and emergency supply distribution.

Target region and data availability

Regions of interest

The region includes:

- Seven districts (e.g., one main city, one town, one or two rural districts).
- Mix of urban centers and surrounding suburban areas.
- The exercise focuses on one month during the flood/cyclone season.

Data for this exercise

- Synthetic Call Detail Records (CDR) from one mobile network operator (MNO) including 1,000 IDs covering one-month period.
- Synthetic census population data (source: WorldPop)
- Administrative boundary data

Case scenario

- You are assessing an OD matrix generated from MPD for disaster preparedness.
- The OD matrix shows daily inter-district mobility between the selected districts.
- Only one MNO provided data for this project.
- The OD flows are interpreted with respect to total census population.

Tasks

Step 1 – Data integrity & coverage

Assess whether the MPD data provides sufficient temporal and spatial coverage for the intended purpose.

Step 2 – Analytical robustness

Compare OD results from two alternative methods and evaluate how sensitive the flows are to methodological choices.

Step 3 – Interpretability & use

Determine how and where the indicator can be used in disaster-related decision-making with high confidence, and where it must be accompanied by strong caveats.

Step I: Data integrity & coverage

Key questions:

- Is the data complete across time for this 1-month disaster-season period?
- Does the data provide enough coverage in each district?
- Are there obvious gaps or anomalies that could undermine the OD estimates?

Step 1: What you will compute...

Daily activity

- Number of active subscribers per day (unique subscribers with at least one record).

District coverage

- Assign each subscriber a home district (district with the highest number of records). For each district:
 - Number of MNO subscribers (home in that district).
 - Coverage ratio = MNO subscribers / census population.

Simple integrity checks

- Days with zero or very low activity.
- Number of subscribers with only a single record.

Step 2: Analytical robustness

Key questions:

- Are OD flows stable across reasonable method choices?
- Which OD pairs are highly sensitive to the method used?
- Does this sensitivity affect the suitability of the indicator for disaster planning?

Setups (used for this exercise):

- We computed two OD pairs in the notebook using 30 days of data for the same sample of users:
 - Method A: Baseline method.
 - Speed threshold: high_speed = 10 km/h & low_speed = 3 km/h
 - Method B: Alternative method with different movement definition.
 - Speed threshold: high_speed = 20 km/h & low_speed = 3 km/h

Step 2: What you will compute...

Merge the Method A and Method B OD tables by origin–destination pair for comparison.

For each OD pair, compute the following values to assess sensitivity to the change in method:

- Flow_A, Flow_B
- Absolute difference = $\text{Flow}_B - \text{Flow}_A$
- Relative difference = $(\text{Flow}_B - \text{Flow}_A) / \text{Flow}_A$ (in %)

Step 3: Interpretability & use

Key questions:

- Given what we know about coverage (Step 1) and robustness (Step 2), how should we interpret the OD matrix for disaster response?
- For which districts or OD pairs is the indicator reliable enough for operational decisions?
- Where are strong caveats or additional data needed?

Step 3: What you will do...

For each district (key OD pair), combine:

- Coverage information (coverage ratio, coverage flag: High / Medium / Low).
- Method sensitivity (maximum relative difference for important flows, robustness flag).

Evaluate an overall quality level (e.g., High / Medium / Low) and propose a recommended use:

- Suitable for detailed evacuation planning.
- Suitable only for rough prioritization.
- Not suitable without additional information.



The GDF-MPD window is a multi-partner window funded by the Spanish Government, Ministry of Economy, Commerce and Business. The GDF-MPD program is jointly managed by DEC (DECDG, DIME), Poverty and Digital Development Global Practices in technical partnership with the International Telecommunications Union (ITU).





Mobile Phone Data for Policy Programme (WB-GDF)
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